

COS 433: Cryptography

Announcements:

- Problem Set 4 due 11/12
- Problem Set 5 to be released 11/12, due 12/3

This week: Zero-Knowledge Proofs

Proofs

- Ax. 1. $(P(\varphi) \wedge \Box \forall x(\varphi(x) \Rightarrow \psi(x))) \Rightarrow P(\psi)$
- Ax. 2. $P(\neg\varphi) \Leftrightarrow \neg P(\varphi)$
- Th. 1. $P(\varphi) \Rightarrow \Diamond \exists x \varphi(x)$
- Df. 1. $G(x) \Leftrightarrow \forall \varphi(P(\varphi) \Rightarrow \varphi(x))$
- Ax. 3. $P(G)$
- Th. 2. $\Diamond \exists x G(x)$
- Df. 2. $\varphi \text{ ess } x \Leftrightarrow \varphi(x) \wedge \forall \psi(\psi(x) \Rightarrow \Box \forall y(\varphi(y) \Rightarrow \psi(y)))$
- Ax. 4. $P(\varphi) \Rightarrow \Box P(\varphi)$
- Th. 3. $G(x) \Rightarrow G \text{ ess } x$
- Df. 3. $E(x) \Leftrightarrow \forall \varphi(\varphi \text{ ess } x \Rightarrow \Box \exists y \varphi(y))$
- Ax. 5. $P(E)$
- Th. 4. $\Box \exists x G(x)$



Proofs

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Traditional/axiomatic proofs

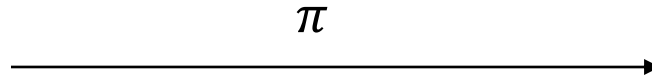


A seminar??

Proofs



Prover



Verifier

CS insight: think of a proof as a form of **computation**

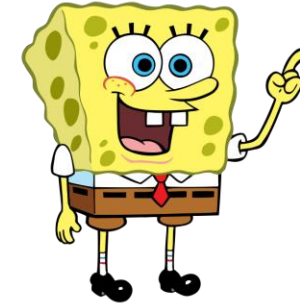
Proofs



Prover

Claim: $\forall x \forall y \forall z (\neq 0),$
 $\forall n > 2, x^n + y^n \neq z^n$

π



Verifier

CS insight: think of a proof as a form of **computation**

E.g.: AW writes a proof in a paper, Bob can then read the paper and check it.

Proofs



Prover

Claim: $\forall x \forall y \forall z (\neq 0),$
 $\forall n > 2, x^n + y^n \neq z^n$

π



Verifier



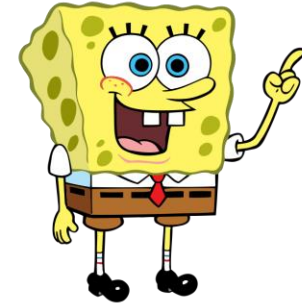
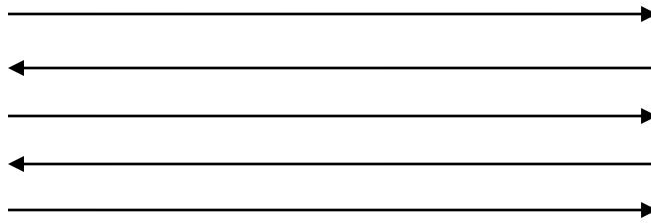
A seminar talk:

- Does not provide an entire, formal proof
- Goal (one of many): convince the audience that a formal proof **exists**
- Leverages audience interaction to make the proof easier to check

Proof Systems



P



V

Definition: a proof system consists of a pair of **interactive algorithms** (P, V)

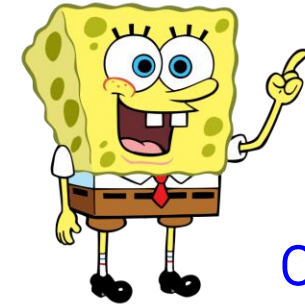
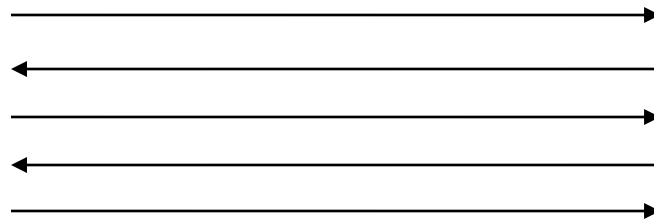
- V will always be a **polynomial-time** algorithm
- Polynomial with respect to what?

Proof Systems

Claim: $x \in L$



$P(x, w)$



$V(x)$

Output: $b \in \{0,1\}$

Definition: a proof system for a language $L \subset \{0,1\}^*$ consists of a pair of interactive algorithms (P, V) run on an input $x \in \{0,1\}^n$

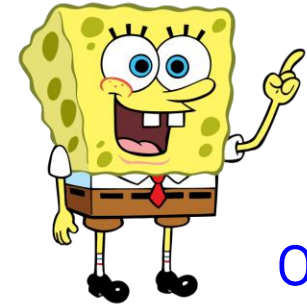
- V will always be a $\text{poly}(n)$ -time algorithm
- P will be either **unbounded**, or polynomial-time given some additional information $w \in \{0,1\}^{\text{poly}(n)}$ (“efficient prover”)

Proof Systems

Claim: $x \in L$



$P(x)$



$V(x)$

Output: $b \in \{0,1\}$

Definition: a proof system for a language $L \subset \{0,1\}^*$ consists of a pair of interactive algorithms (P, V) run on an input $x \in \{0,1\}^n$

- V will always be a **poly**(n)-time algorithm

It must satisfy the following properties:

1. **Completeness:** if $x \in L$ and the two parties run (P, V) , then V outputs 1 (accept)
2. **Soundness:** if $x \notin L$, then for **any** strategy P^* , V accepts with probability $\text{negl}(n)$.

Note: we allow for P, V to be **randomized**.

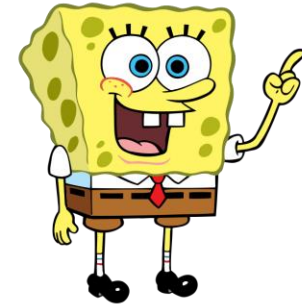
Proof Systems: Examples

Claim: $N \in L$



$P(N)$

(c, d)



$V(N)$

Check that $c, d > 1$ and $N = cd$

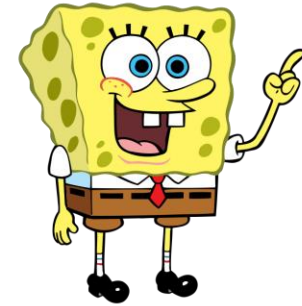
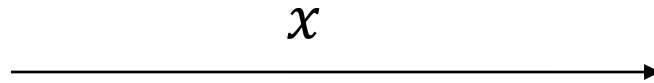
Example: composite integers

- $L = \{N: N \text{ is composite}\}$
- $P(N)$: factor $N = c \cdot d$, send (c, d)
- $V(N, c, d)$: check that $N = cd$ and $c, d > 1$

Use case: Alice could generate and release $N = pq$, and prove that it is composite at any later time by sending (p, q)

Proof Systems: Examples

Claim: $(N, y) \in L$ $\iff$ $P(N, y)$



$V(N, y)$

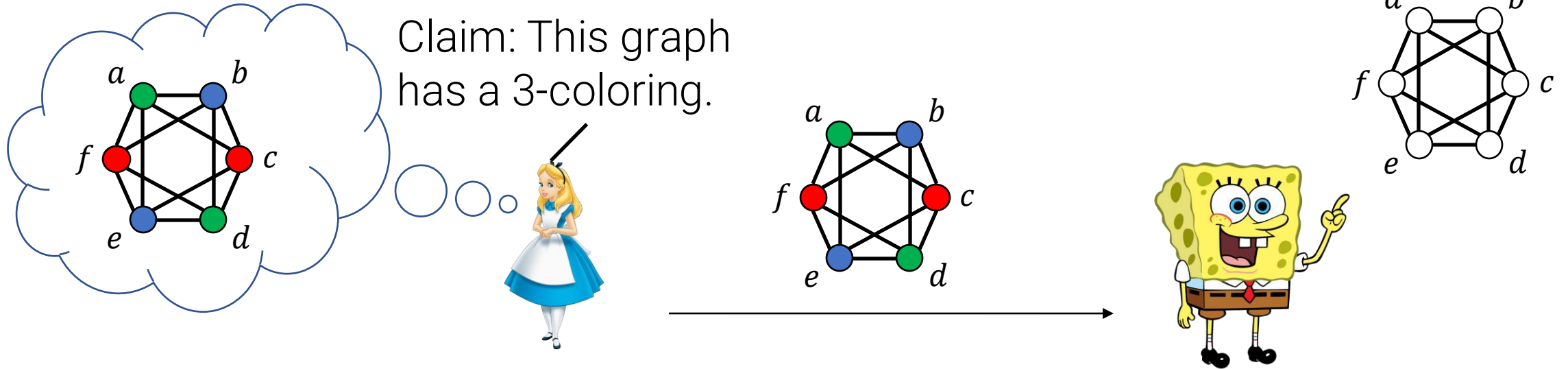
Check that $y \equiv x^2 \pmod{N}$

Example: quadratic residuosity

- $L = \{(N, y): y \text{ is a square mod } N\}$
- $P(N)$: find and send square root x
- $V(N, y, x)$: check that $y \equiv x^2 \pmod{N}$

Use case: Alice could generate and release $N = pq$, and (efficiently) prove that any square y is a perfect square (root finding is easy given factorization).

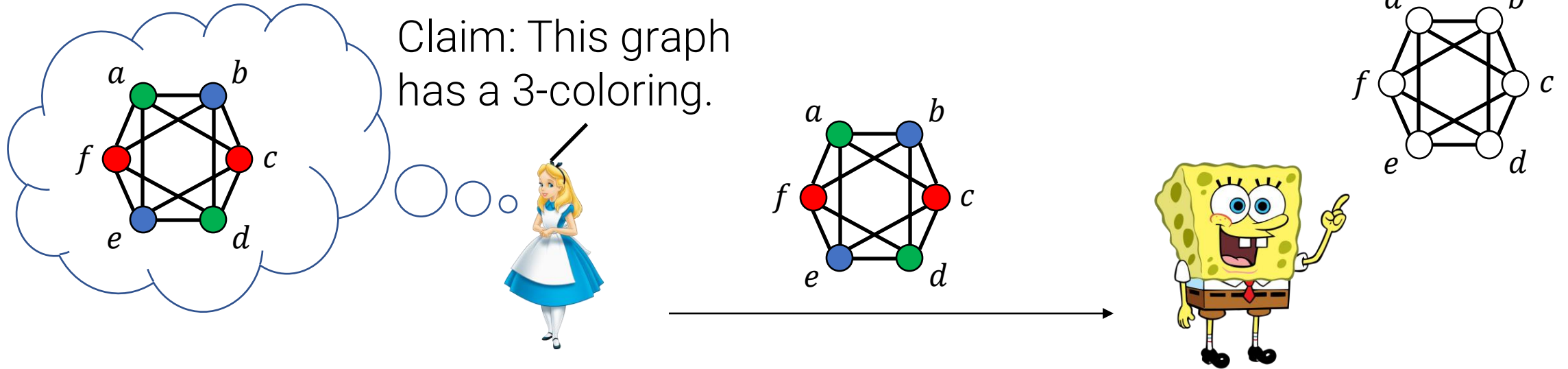
Proof Systems: Examples



Example: graph 3-coloring

- $L = \{\text{graph } G: G \text{ is 3-colorable}\}$
- $P(G)$: find and send color assignment $c: \text{Vert}(G) \rightarrow \{0, 1, 2\}$
- $V(G, c)$: check that all edges of G are bichromatic
 - Takes time $O(n^2)$ in the worst case

Proof Systems: Examples



These are all extremely simple proof systems!

- Non-interactive
- Deterministic

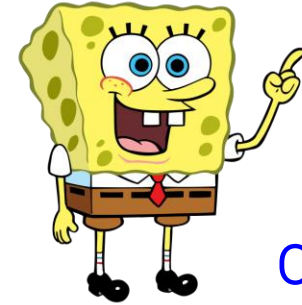
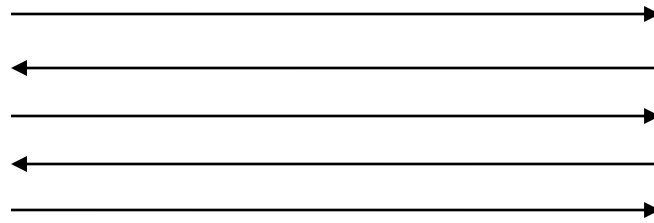
Definition: The complexity class NP is the set of all languages $L \subset \{0,1\}^*$ that have a deterministic, non-interactive proof system

Proof Systems: Examples

Claim: $x \in L$



$P(x, w)$



$V(x)$

Output: $b \in \{0,1\}$

Today and next time: focus on proof systems for languages $L \in \text{NP}$

- For $x \in L$ and non-interactive proof system for L , call a valid proof a *witness* w
- Assume honest prover has a witness w .

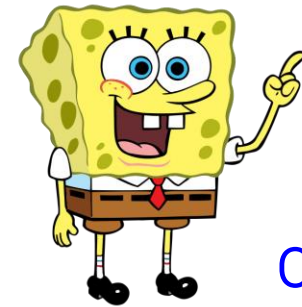
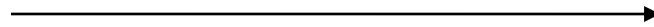
Proof Systems: Examples

Claim: $x \in L$



$P(x, w)$

w



$V(x)$

Output: $b \in \{0,1\}$

Today and next time: focus on proof systems for languages $L \in \text{NP}$

- For $x \in L$ and non-interactive proof system for L , call a valid proof a *witness* w
- Assume honest prover has a witness w .
- Trivial proof system: P sends w , V checks that w is valid.
- We've done nothing...?

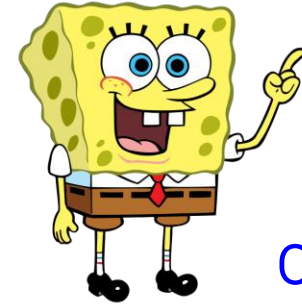
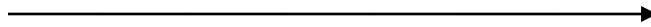
Proof Systems

Claim: $x \in L$



$P(x, w)$

w



$V(x)$

Output: $b \in \{0,1\}$

Where is the **cryptography**?

Idea: maybe w is supposed to be a **secret**!

- w could be Alice's secret key, x could be an associated public key
- w could contain sensitive data that Alice doesn't want to (or legally cannot) release

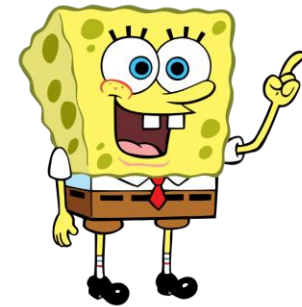
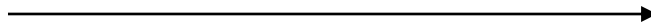
Proof Systems: Quadratic Non-Residuosity

Claim: $y \notin \text{QR}_N$



$P(p, q, y)$

p, q



$V(N, y)$

Check $N = pq$, check if y is a square.

Can we prove y is not a square, **without revealing** the factorization of N ?

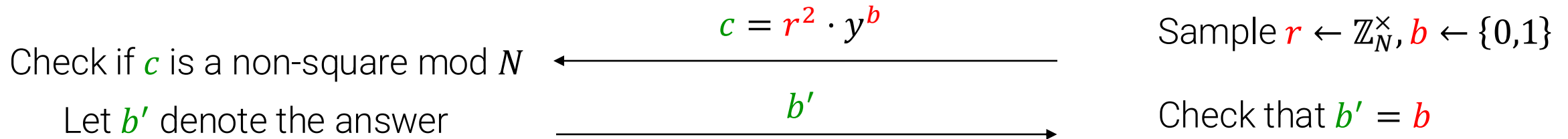
- **Problem:** verifier will not be able to decide if $y \in \text{QR}_N$ on his own.
- **Solution:** leverage **interaction** and **randomness**

Proof Systems: Quadratic Non-Residuosity

$P(p, q, y)$

Claim: $y \notin \text{QR}_N$

$V(N, y)$



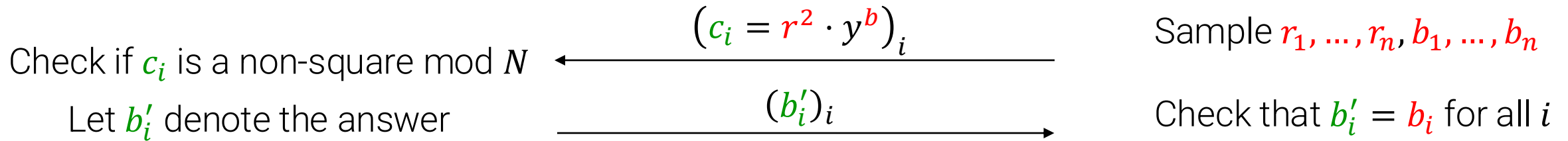
- **Completeness:** if y is not a square, then c is a non-square if and only if $b = 1$
- **Soundness:** if y is a square,
 - c is a random square mod N
 - so c hides the value of b
 - P^* can only output $b' = b$ with probability $\frac{1}{2}$
 - “Soundness error $\frac{1}{2}$ ”

Proof Systems: Quadratic Non-Residuosity

$P(p, q, y)$

Claim: $y \notin \text{QR}_N$

$V(N, y)$



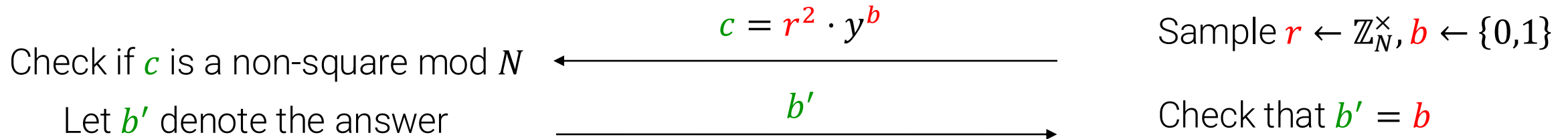
- **Soundness:** P^* can only output all $b'_i = b_i$ with probability 2^{-n} .

Proof Systems: Quadratic Non-Residuosity

$P(p, q, y)$

Claim: $y \notin \text{QR}_N$

$V(N, y)$



Question: can V learn the factorization of N after receiving a proof?

- Let's assume that V follows the protocol

Answer: no (assuming factoring N is hard), and we can prove this!

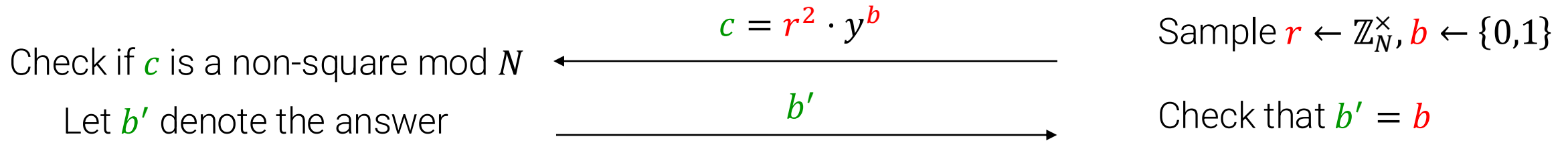
- Suppose V could output (p, q) additionally given (r, b, c, b') (where r is red, b is red, c is green, and b' is green)
- Algorithm $V(N, y, r, b, c, b') \rightarrow (p, q)$
- Then we can efficiently factor N : run $V(N, y, r, b, c, b)$ (where r is red, b is red, and c is green)

Proof Systems: Quadratic Non-Residuosity

$P(p, q, y)$

Claim: $y \notin \text{QR}_N$

$V(N, y)$



Amazing property: V seems to learn **nothing** by receiving a proof!

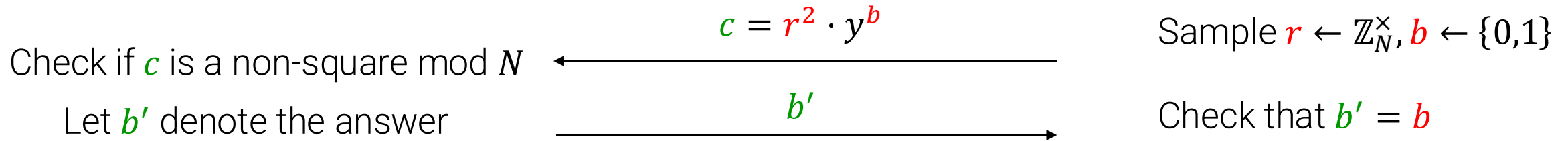
- V does learn that y is a non-square
- But that's it!

Proof Systems: Quadratic Non-Residuosity

$P(p, q, y)$

Claim: $y \notin \text{QR}_N$

$V(N, y)$



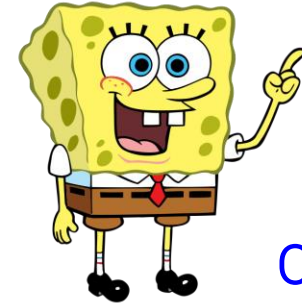
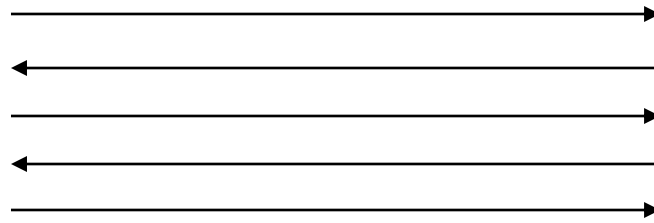
We formalize this via a security property called (Honest-Verifier) Zero Knowledge

Proof Systems: Examples

Claim: $x \in L$



$P(x, w)$



$V(x)$

Output: $b \in \{0,1\}$

Definition: given a pair of interactive algorithms $P(x, w) \leftrightarrow V(x)$, we define the **view of V** , $\text{view}_V(x, w)$, to consist of:

- The input x
- All random coins sampled by V , and
- All messages sent from P to V in the protocol

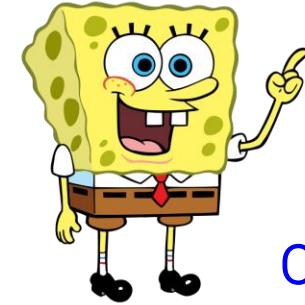
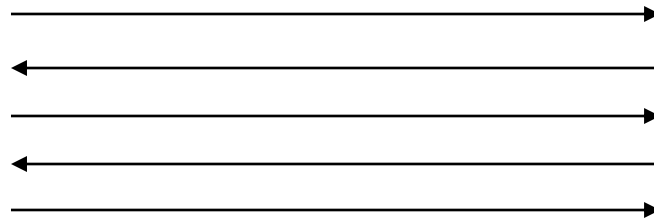
view_V is a distribution w.r.t. the randomness of P and V

Proof Systems: Examples

Claim: $x \in L$



$P(x, w)$



$V(x)$

Output: $b \in \{0,1\}$

Definition: a proof system (P, V) for a language L is **honest-verifier zero knowledge** if there exists a **polynomial-time simulator**

$$\text{Sim}(x) \rightarrow \tau$$

such that **for all $x \in L$** and all valid w ,

$$\text{Sim}(x) \approx_c \text{view}_V(x, w)$$

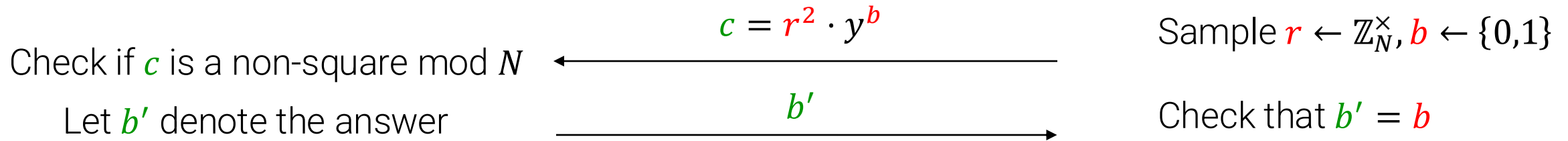
Simulation paradigm: everything V learned, V could have generated on their own!

Proof Systems: Quadratic Non-Residuosity

$P(p, q, y)$

Claim: $y \notin \text{QR}_N$

$V(N, y)$



Claim: this proof system is HVZK

Proof: define the simulator

$$\text{Sim}(N, y) = ((N, y), (r, b), b)$$

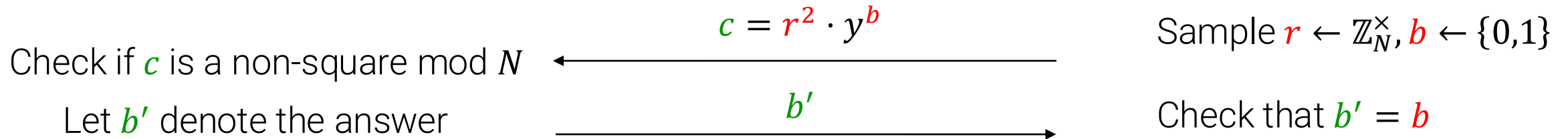
If y is a non-square, the output of $\text{Sim}(N, y)$ is identical to $\text{view}_V(N, y, p, q)$

Proof Systems: Quadratic Non-Residuosity

$P(p, q, y)$

Claim: $y \notin \text{QR}_N$

$V(N, y)$



Problem: what if we don't trust V to behave according to the protocol specification?

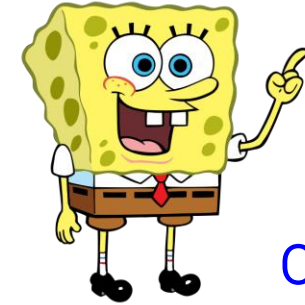
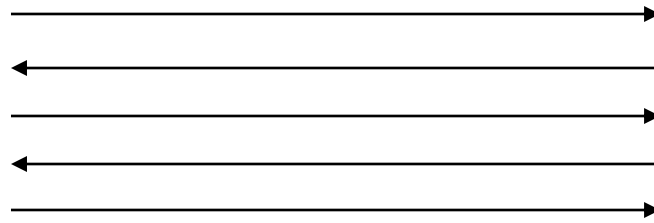
Continue next time

Proof Systems: Examples

Claim: $x \in L$



$P(x, w)$



$V(x)$

Output: $b \in \{0,1\}$

Definition: a proof system (P, V) for a language L is malicious verifier zero knowledge if for all polynomial-time V^* , there exists a polynomial-time simulator

$$\text{Sim}^*(x) \rightarrow \tau$$

such that for all $x \in L$ and all valid w ,

$$\text{Sim}^*(x) \approx_c \text{view}_{V^*}(x, w)$$

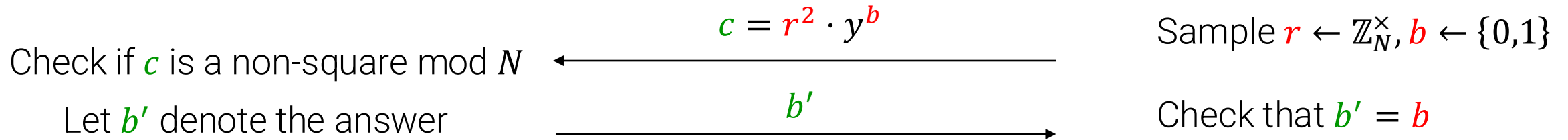
Simulation paradigm: everything V^* learned, V^* could have generated on their own!

Proof Systems: Quadratic Non-Residuosity

$P(p, q, y)$

Claim: $y \notin \text{QR}_N$

$V(N, y)$



Is this protocol malicious-verifier ZK?

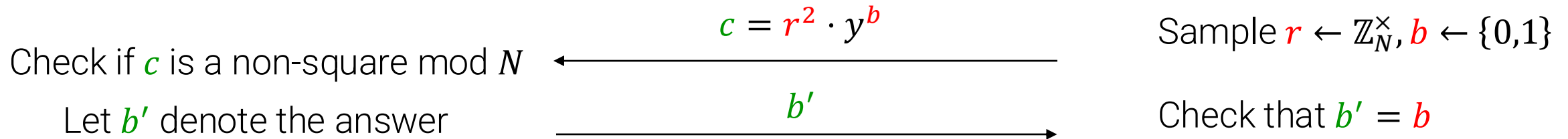
No!

Proof Systems: Quadratic Non-Residuosity

$P(p, q, y)$

Claim: $y \notin \text{QR}_N$

$V(N, y)$



Is this protocol malicious-verifier ZK?

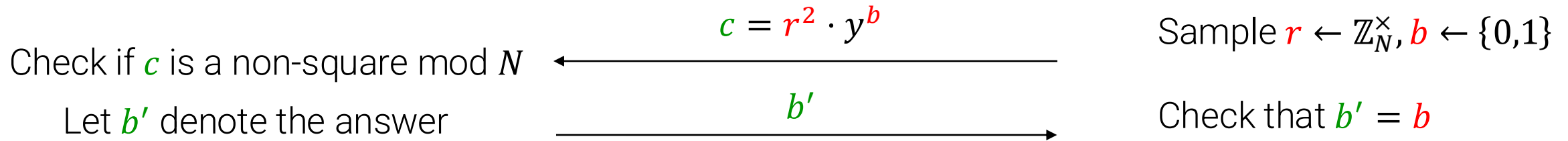
- Intuition: malicious V^* could send c to be any number mod N , where V^* does not know if c is a square modulo N
- Honest P will tell V^* this information, so V^* has learned something

Proof Systems: Quadratic Non-Residuosity

$P(p, q, y)$

Claim: $y \notin \text{QR}_N$

$V(N, y)$



Magic: there's a way to change this protocol, to make it fully ZK!

A little too complicated; we will switch to a different example.

Zero-Knowledge Proof for QR

$P(N, y, w)$

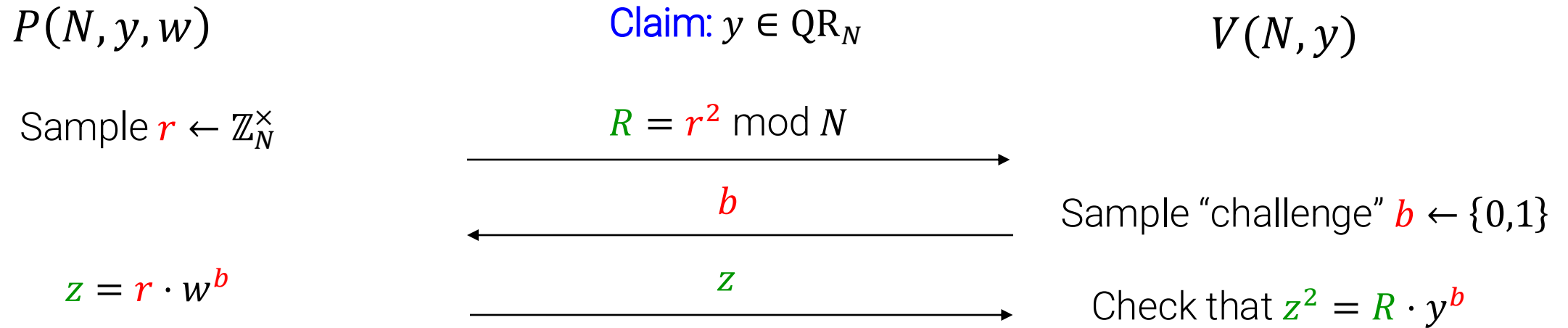
Claim: $y \in \text{QR}_N$

$V(N, y)$

Example: quadratic residuosity

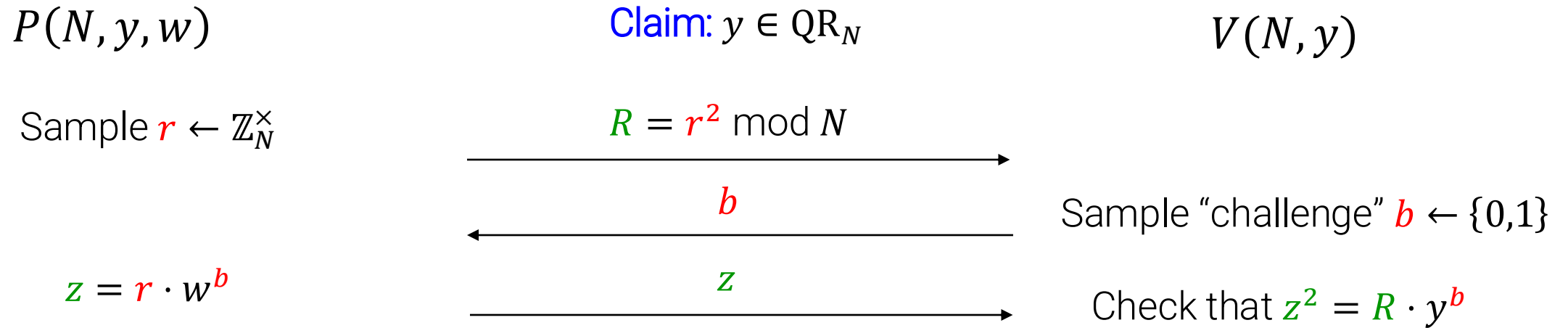
- $L = \{(N, y): y \text{ is a square mod } N\}$
- Witness w : $w^2 \equiv y \pmod{N}$

Zero-Knowledge Proof for QR



Intuition: “I’ll show you that I could send you a square root if I wanted... but I won’t!”

Zero-Knowledge Proof for QR

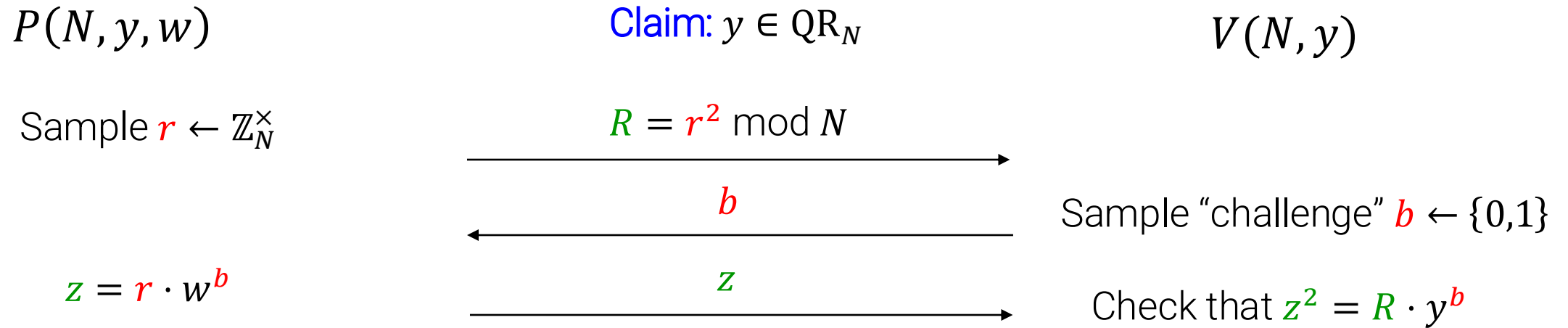


Completeness: $(r \cdot w^b)^2 \equiv R \cdot y^b$

Soundness: if y is **not** a square, then **impossible** to have answers to both challenges!

- For any $R \in \mathbb{Z}_N^\times$, either R is not a square or $R \cdot y$ is not a square
- For any P^* , V rejects with probability at least $\frac{1}{2}$
- Repeat the protocol (sequentially!) to improve the soundness error.

Zero-Knowledge Proof for QR



Claim: this proof system is HVZK

Proof: define the simulator

$$\text{Sim}(N, y) = \left((N, y), \overbrace{b, s^2 \cdot y^{-b}}^R, \underbrace{s}_z \right)$$

If $y \equiv w^2$, the output of $\text{Sim}(N, y)$ is **identical** to $\text{view}_V(N, y, w)$

Zero-Knowledge Proof for QR

$P(N, y, w)$

Claim: $y \in \text{QR}_N$

$V(N, y)$

Sample $r \leftarrow \mathbb{Z}_N^\times$

$$R = r^2 \bmod N$$

b

Sample "challenge" $b \leftarrow \{0,1\}$

$$z = r \cdot w^b$$

z

Check that $z^2 = R \cdot y^b$

Claim: this proof system is ZK

Proof: ???

Any ideas?

Continue next time